

Abstract

Kernel mean embedding is a useful tool to represent and compare probability measures. Despite its usefulness, kernel mean embedding considers infinite-dimensional features, which are challenging to handle in the context of differentially private data generation. A recent work (Harder et al., 2021) proposes to approximate the kernel mean embedding of data distribution using finite-dimensional random features, which yields analytically tractable sensitivity. However, the number of required random features is excessively high, which worsens the privacy-accuracy trade-off. To improve the trade-off, we propose to replace random features with Hermite polynomial features. Unlike the random features, the Hermite polynomial features are ordered, where the features at the low orders contain more information on the distribution than those at the high orders. Hence, a relatively low order of Hermite polynomial features can more accurately approximate the mean embedding of the data distribution compared to a significantly higher number of random features.

Differential Privacy

A randomized algorithm, $\mathcal{M}$, is said to be (ϵ, δ) -DP, if for all possible sets of outputs S and for all neighboring datasets $\mathcal{D}, \mathcal{D}'$ differing in an only single entry ($d(\mathcal{D}, \mathcal{D}') \leq 1$) we have:

$$\Pr[\mathcal{M}(\mathcal{D}) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(\mathcal{D}') \in S] + \delta$$

A method of preserving (ϵ, δ) -DP is **Gaussian Mechanism**:

$$\mathcal{M}(\mathcal{D}) = f(\mathcal{D}) + \mathcal{N}(0, \sigma^2 \Delta_f^2 \mathbf{I}_d),$$

for sensitivity $\Delta_f = \max_{\mathcal{D}, \mathcal{D}', d(\mathcal{D}, \mathcal{D}') \leq 1} \|f(\mathcal{D}) - f(\mathcal{D}')\|_2$ and $\sigma = \sqrt{\frac{2 \log \frac{1.25}{\delta}}{\epsilon^2}}$.

Maximum Mean Discrepancy

Definition: $\text{MMD}^2(P, Q) = \left\| \mathbb{E}_{x \sim P}[\phi(x)] - \mathbb{E}_{y \sim Q}[\phi(y)] \right\|_{\mathcal{H}}^2$

- P and Q are two probability measures
- $\phi: \mathcal{X} \rightarrow \mathcal{H}$ is a feature map such that, $k(x, y) = \langle \phi(x), \phi(y) \rangle$ for a positive definite kernel k
- If k is a **characteristic kernel**: $\text{MMD}(P, Q) = 0$ iff. $P = Q$.
- **Kernel approximation**: approximate $k(x, x')$ with **finite dimensional** feature vectors $\hat{\phi}(x)$ such that

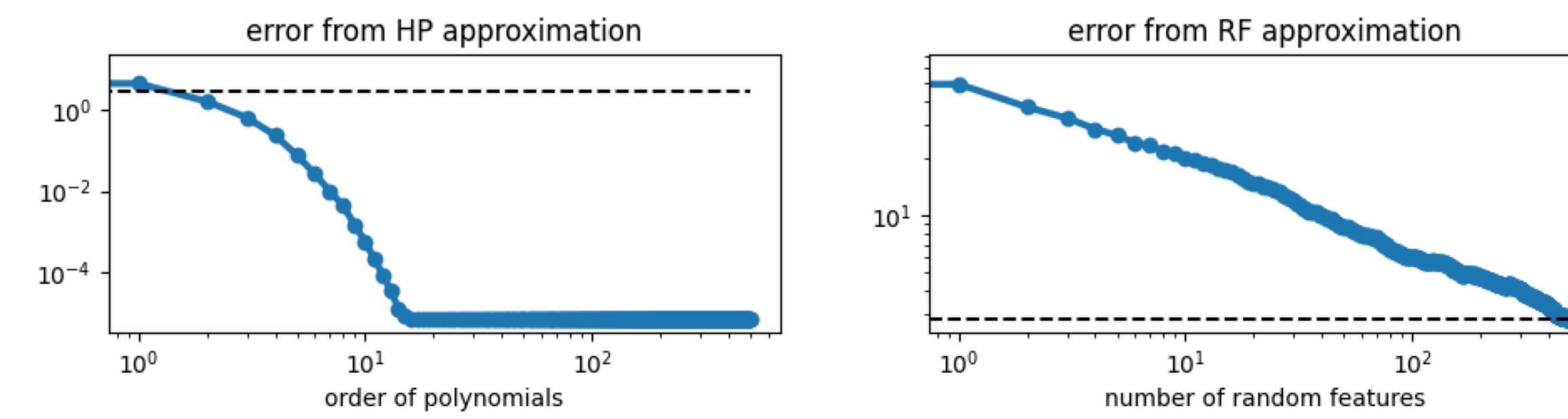
$$\widehat{\text{MMD}}^2(P, Q) = \left\| \mathbb{E}_{x \sim P}[\hat{\phi}(x)] - \mathbb{E}_{y \sim Q}[\hat{\phi}(y)] \right\|_2^2 \simeq \text{MMD}(P, Q)$$

Hermite polynomial

We can approximate the **Gaussian kernel** via **Hermite polynomial features** $\hat{\phi}_{HP}^{(C)}(x) = [\sqrt{\lambda_0}f_0(x), \dots, \sqrt{\lambda_C}f_C(x)]$ where $\lambda_i > 0$ and $f_i(x) = H_i(x)\exp(-\rho/(1+\rho)x^2)/\sqrt{N_i}$ represent the eigenvalues and eigenfunctions of the kernel. Here, H_i denotes **Physicists' Hermite polynomial of order c**:

$$H_c(x) = (-1)^c \exp(x^2) \frac{d^c}{dx^c} \exp(-x^2)$$

- **Lower orders of Hermite features contain more information of the distribution!**



- Hermite polynomial features for the outer product kernel has size $(C+1)^D$ where C is the Hermite polynomial order and D is data dimension $\Rightarrow$ **computational prohibitive!**

Our suggestion: combined kernel

Definition: $k_c(x, x') = k_p(x, x') + k_s(x, x')$,

where $k_p(x, x')$ is the approximated **outer product kernel** over some subsampled dimensions ($D_{prod} \ll D$) and $k_s(x, x')$ is the **sum kernel** over input data dimension.

- D_{prod} relatively small needed for computational tractability
- Use of sum kernel to provide extra information about the unselected input dimensions in the outer product kernel.

Our algorithm: DP-HP

Goal: Match real data distribution P with the private generated Q by $\min_{\theta} \gamma \|\hat{\mu}_P^p - \hat{\mu}_{Q_\theta}^p\|_2^2 + \|\hat{\mu}_P^s - \hat{\mu}_{Q_\theta}^s\|_2^2$ where γ is a positive constant that helps dealing with the scale difference between the two terms.

- Only $\hat{\mu}_P^p$ and $\hat{\mu}_P^s$ depend on real data.
 - $\hat{\mu}_P^s$ privatized once
 - $\hat{\mu}_P^p$ privatized each time we subsample dimensions
- Hermite polynomial features are **norm bounded**: $\|\phi_{HP,d}^{(C)}(x_d)\|_2^2 \leq 1$ for $d = 1, \dots, D$
- Approximated mean embeddings have **sensitivity** inversely proportional to the number of data samples: $\Delta_{\mu_P} = \frac{2}{m}$

Algorithm 1 DP-HP

Require: Dataset $\mathcal{D}$, privacy budget (ϵ, δ) , number of outer product subsampling dimension D_{prod} , scale parameter γ , number of epochs T

Step 1. Compute privacy parameter σ based on (ϵ, δ)

Step 2. Release mean embedding $\hat{\mu}_P^s$ with Gaussian mechanism.

for $t = 1, \dots, T$ **do**

Step 3. Subsample D_{prod} dimension from D and release mean embedding $\hat{\mu}_P^p$ with Gaussian mechanism.

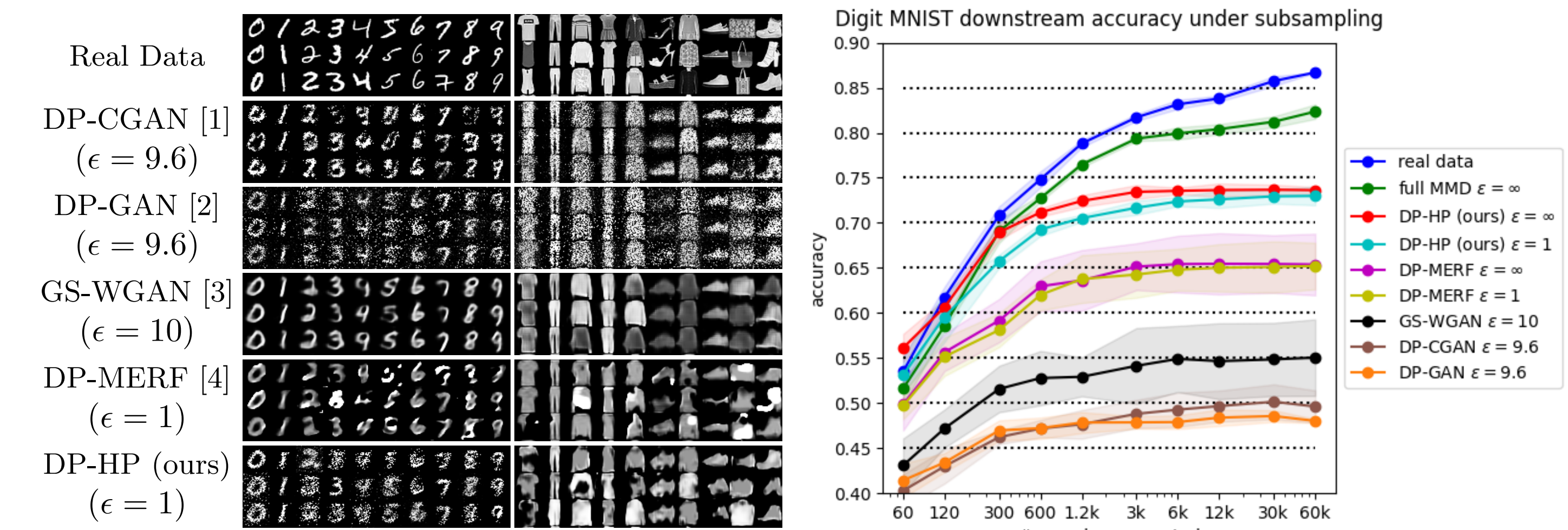
Step 4. Train generator by $\min_{\theta} \gamma \|\hat{\mu}_P^p - \hat{\mu}_{Q_\theta}^p\|_2^2 + \|\hat{\mu}_P^s - \hat{\mu}_{Q_\theta}^s\|_2^2$

end for

Step 5. Sample synthetic dataset from generator.

Experiments: MNIST

DP-HP outperforms other state-of-the-art methods for private synthetic data generation



TAKEAWAYS:

- DP-HP effective data size (no significant improvement on test accuracy when synthetic sample size increases) is comparable to other DP methods considered and lies about 5-10% (3k-6k) of the original data size
- For the same privacy level, DP-HP outperforms DP-MERF in terms of accuracy by 10% on MNIST.
- Compared with GAN models, DP-HP outperforms all of them with a tighter privacy constraint.

Conclusions

- The use of Hermite polynomial features **improves the accuracy-privacy trade-off** thanks to the orderedness of the polynomial features.
- The combined kernel enables **computational tractability** in handling high-dimensional data.
- DP-HP outperforms other state-of-the-art methods for synthetic private data generation